Case Study 4: Advanced Numeric Pattern Generator

Assignment Case No. 27 | Level: Beginner

01 PROBLEM STATEMENT

Produce a hollow diamond or symmetric numerical pattern based on user-specified dimensions, with validation for invalid dimensions and carefully aligned output.

Problem interpretation: Generate a symmetric hollow diamond using a positive integer size value. The program validates the size and prints stars only on the boundary of the diamond.

02 OBJECTIVES

Use nested loops to generate a pattern.

Validate the requested size.

Print a symmetric hollow diamond.

Understand how row and column positions control output.

03 ASSUMPTIONS

The size is a positive integer.

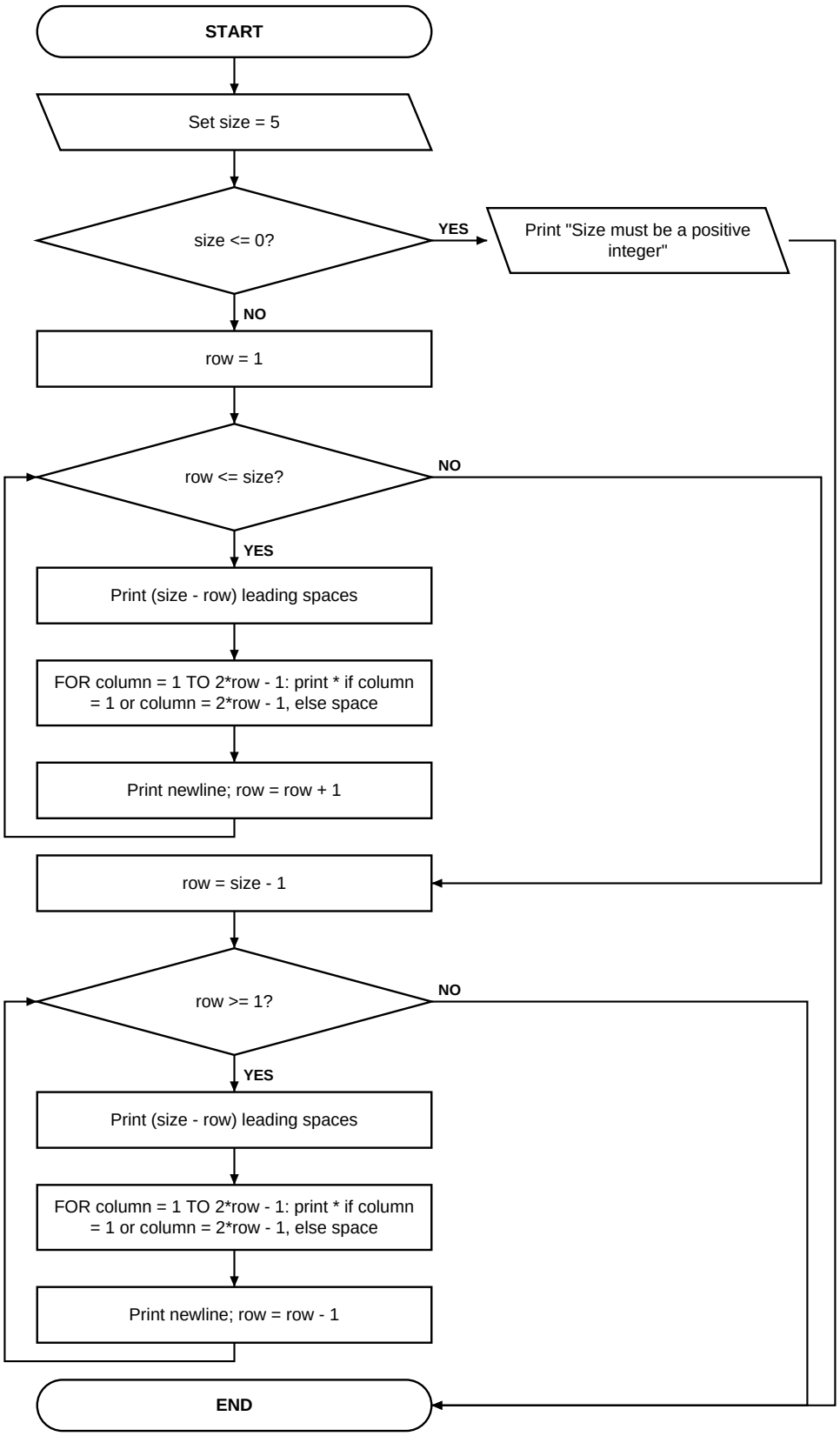
The example uses size 5 for the upper and lower halves, making a 9-row diamond.

A star is printed at the left and right boundary; interior positions are spaces.

04 ALGORITHM

1. Start.
2. Set size to 5.
3. If size is not positive, print an error.
4. Print the upper half including its middle row using nested loops.
5. Print the lower half using nested loops.
6. Stop.

05 FLOWCHART (TEXT FLOW)



06 PSEUDOCODE

```
BEGIN
    size = 5
    IF size <= 0 THEN
        PRINT "Size must be a positive integer"
    ELSE
        FOR row = 1 TO size
            print (size - row) spaces
            FOR column = 1 TO 2*row - 1
                IF column = 1 OR column = 2*row - 1 THEN print "*"
                ELSE print " "
                END IF
            END FOR
            print newline
        END FOR
        FOR row = size - 1 DOWN TO 1
            print (size - row) spaces
            FOR column = 1 TO 2*row - 1
                IF column = 1 OR column = 2*row - 1 THEN print "*"
                ELSE print " "
                END IF
            END FOR
            print newline
        END FOR
    END IF
END
```

07 PYTHON SOURCE CODE

The complete Python program is provided in the separate file 04_Advanced_Numeric_Pattern_Generator_Code.pdf.

08 TEST CASES

INPUT / SCENARIO	CHECK	EXPECTED RESULT
size = 5	9-row hollow diamond	Valid; stars only on the boundary
size = 1	Single row	One star
size = 0	Size not positive	Size must be a positive integer
size = -3	Size not positive	Size must be a positive integer

09 SAMPLE OUTPUT PREVIEW

```
    *
   * *
  *   *
 *     *
*       *
 *     *
  *   *
   * *
    *
```

10 CONCLUSION

The pattern generator demonstrates nested loops, conditions, string repetition, and alignment through spaces.